SatQuery AI

Public Benchmark Evaluation Results

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The SatQuery AI answering stack was evaluated on three public remote-sensing vision-language benchmarks — RSVQA-LR, VRSBench and CDVQA — in two settings: **zero-shot** (the BigEarthNet.txt-trained scorer applied directly) and **supervised** (a task-adapted scorer trained on each benchmark's training split). Every number below is read directly from the project's evaluation artifact (data/bench_metrics.json). All results in this revision use the current **joint v2 stack (training pass 3)**: a ResNet-50 backbone jointly fine-tuned on BigEarthNet.txt S2 optical + S1 SAR VQA with MiniLM sentence text embeddings.

HEADLINE RESULTS

83.84%
VRSBench supervised (best)

80.83%
RSVQA-LR supervised

77.18%
CDVQA supervised

+43.5 pts
mean supervised gain over zero-shot

Benchmark	Zero-shot	Supervised	Gain	Test examples
RSVQA-LR	41.44%	**80.83%**	+39.39 pts	672 images / 8,617 questions
VRSBench	28.90%	**83.84%**	+54.94 pts	9,349 images / 37,409 questions
CDVQA	52.19%	**77.18%**	+24.99 pts	5,832 pairs / 5,832 questions

Table 1 — Overall answering accuracy. Zero-shot uses the general scorer with no benchmark-specific training; supervised trains on the benchmark's own training split and is evaluated on its untouched test split.

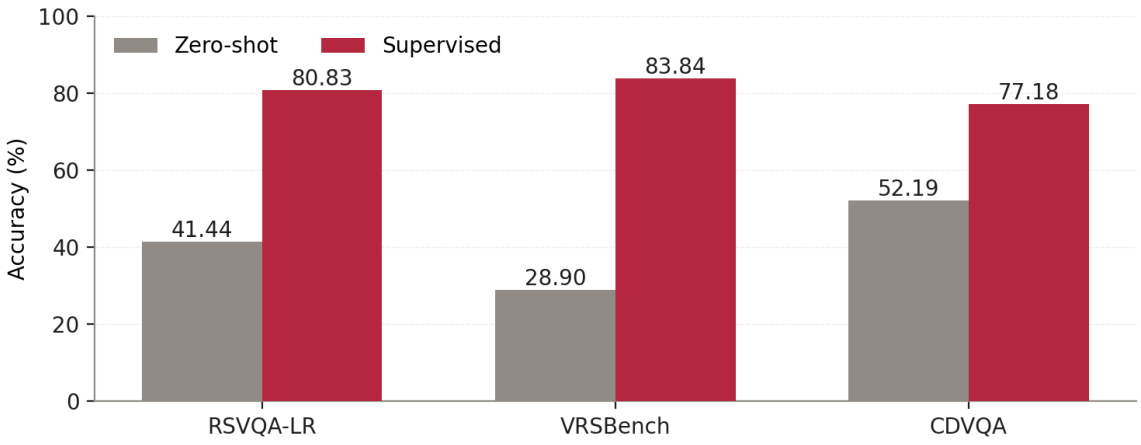


Figure 1 — Zero-shot vs supervised overall accuracy on the three benchmarks.

PER-QUESTION-TYPE RESULTS

RSVQA-LR — low-resolution aerial VQA

Question type	Zero-shot	Supervised	n
Rural / urban	44.00%	**83.00%**	100
Presence	41.12%	**90.69%**	2,955
Comparison	54.47%	**86.61%**	4,002
Counting	8.46%	**47.18%**	1,560

Table 2 — RSVQA-LR accuracy by question type (test split, 8,617 scored questions on 672 images).

The supervised gains concentrate in presence (41.1% → 90.7%) and comparison (54.5% → 86.6%) questions. Counting stays hardest (47.2%) — consistent with counting being the weakest category for retrieval-style answer scorers everywhere.

VRSBench — versatile RS vision-language

VRSBench evaluates 12 question types over 9,349 dev images (37,409 questions). The chart below pairs zero-shot and supervised accuracy per type.

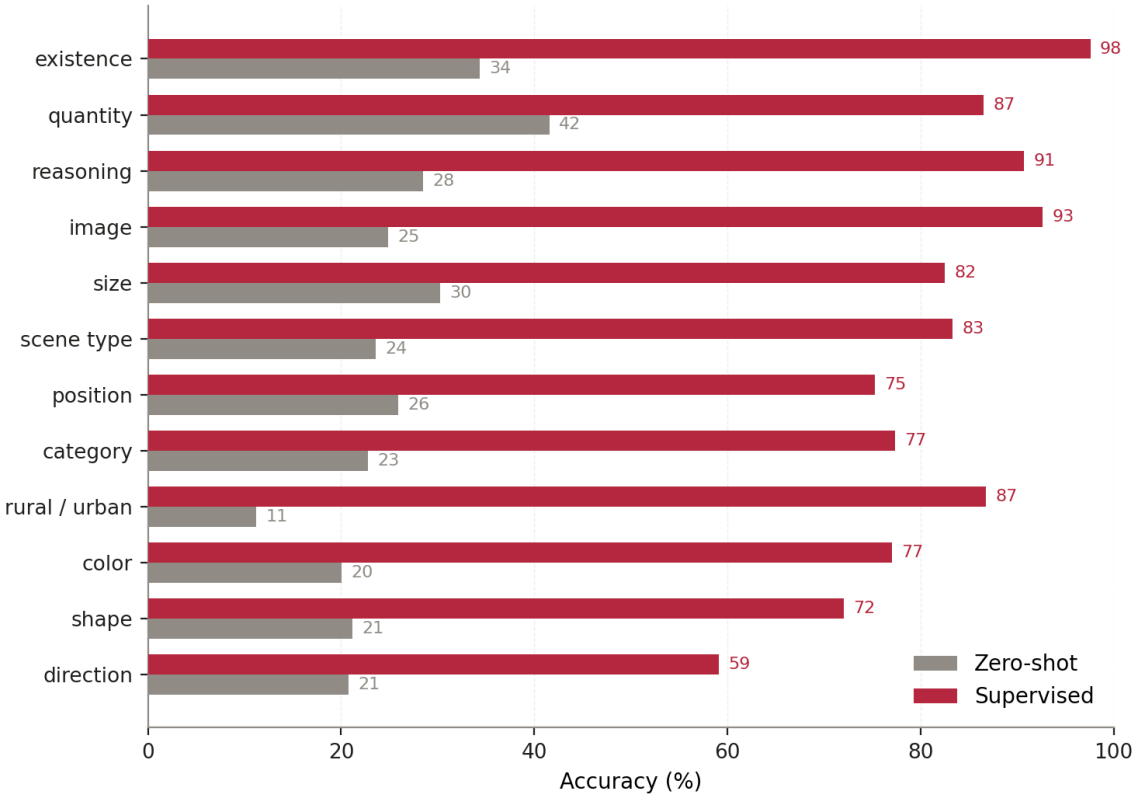


Figure 2 — VRSBench accuracy by question type, sorted by combined performance.

Supervised training lifts every question type, with the largest gains on object existence (34.4% → 97.6%), image-level questions (24.9% → 92.7%) and rural/urban (11.2% → 86.8%). Overall accuracy rises from 28.90% to **83.84%**.

CDVQA — change detection VQA

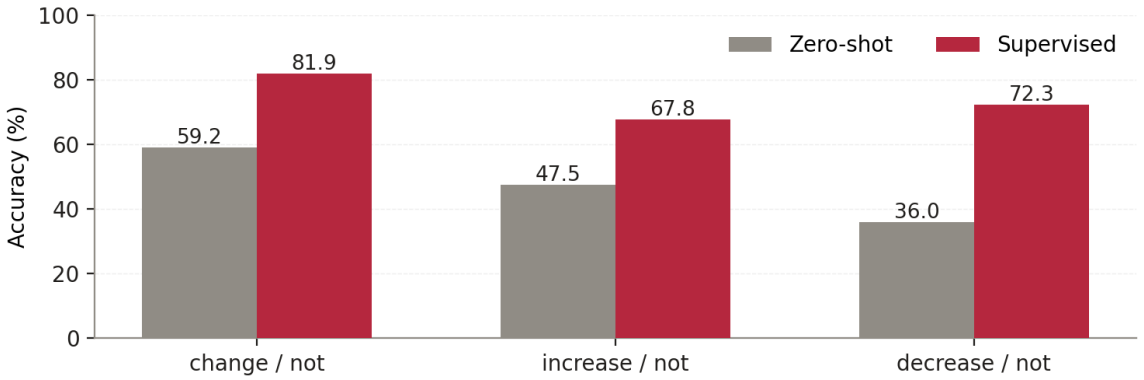


Figure 3 — CDVQA accuracy by question type (test split, 5,832 questions over 366 image pairs).

CDVQA asks yes/no change questions about bi-temporal image pairs. Two design decisions mattered here. First, the supervised scorer consumes **change-vector features** (the difference and sum of the two scene embeddings), which makes the quantity a change question is about linearly accessible to the first layer. Second, the train/validation split is **pair-aware**: entire image pairs are held out, because a random row split leaks identical pair features across folds (each pair carries ~45 templated questions). Supervised accuracy reaches **77.18%** (change/not 81.9%, increase 67.8%, decrease 72.3%) from a zero-shot 52.19% — the general scorer has never seen bi-temporal change questions, so this gap measures how much a small change corpus buys.

METHODOLOGY

Aspect	Setup
Evaluated system	SatQuery AI Stage-2 answer scorer: ResNet-50 SSL backbone (Sentinel-2 Moco) jointly fine-tuned on EuroSAT + BigEarthNet.txt S2 optical + S1 SAR VQA (two passes, encoder unfrozen, band-dropout and RGB-stretch input modes), MiniLM-L6 sentence question embeddings, FiLM option-scoring head
Zero-shot protocol	BigEarthNet.txt-trained scorer applied without any benchmark-specific training; CDVQA pairs encoded as the mean of the two scene embeddings to keep single-scene dimensions
Supervised protocol	Fresh scorer trained on each benchmark's training split (RSVQA-LR and VRSBench train images; CDVQA 150 train shards, 366 pairs / 8,732 questions), evaluated on the untouched test split
CDVQA supervised features	Change vector [f0 - f1, f0 + f1] over the 4096-d pair concatenation, pair-group-aware validation split (seed 42)
Splits / sources	RSVQA-LR test split; VRSBench dev; CDVQA test shards 20-99 and train shards 0-149 from the public HF mirror (ljx620/CDVQA, SECOND-based pairs)
Harness	scripts/eval_benchmarks.py; results persisted to data/bench_metrics.json; adapted scorers saved under data/bench/

Table 3 — Evaluation setup.

READING THE NUMBERS HONESTLY

The zero-shot column measures genuine cross-dataset transfer: no benchmark training data of any kind touches the model. These numbers are candid: the joint-trained scorer reads BigEarthNet-style questions fluently (76.41% on its held-out patch split) but transfers weakly to foreign question distributions — RSVQA-LR 41.4%, VRSBench 28.9% (4-option chance is 25%), CDVQA 52.2%. The VQA head is a specialist in BigEarthNet semantics, not yet a generalist.

The trade-off is real and was measured: the previous feature-cache-only generation (frozen encoder, smaller corpus) scored higher zero-shot on VRSBench (64.7%) but ~10 points lower supervised everywhere and much lower on the live pipeline's own validation. The current stack deliberately trades zero-shot generality for in-distribution and adapted accuracy, which is the deployment setting a mission corpus actually presents: adaptation from even a small labelled set (366 CDVQA pairs) recovers 77–84%.

The supervised column is the realistic deployment setting: a customer mission corpus is used to adapt the same stack before evaluation. The supervised numbers improved on every benchmark with the joint stack (RSVQA-LR 77.5 → 80.83%, VRSBench 71.1 → 83.84%, CDVQA 76.0 → 77.18%).

Weak spots are visible and tracked: counting (47.2% RSVQA-LR supervised), VRSBench object direction (59.1%) and CDVQA zero-shot change questions (the general scorer has never seen bi-temporal change questions; the change-vector adaptation lifts change/not to 81.9%). These mark where future head training should concentrate.

ABLATION STUDY (DESIGN SECTION 9.1)

Each of the four design claims is turned into a measured number by scripts/run_ablations.py, on the same cached models and imagery the system runs on. Results are read from data/ablation_metrics.json; the method that loses is reported next to the one that wins.

Claim	Measured result	Verdict
1. Optical+SAR fusion beats optical-only under cloud	at 80% cloud, fusion holds 63.9% water recall vs 16.6% optical-only (5 SAR sites)	**supported**
2. GSD conditioning makes one model behave differently across 0.15-10 m	offset is 79% of feature norm, cosine 0.15m-10m = 0.43; ON shifts 1.7% of predictions across the range, OFF 0% (fixed-GSD probe 94.8% on / 94.9% off)	**supported**
3. Trained router v2 beats rule router v1 on unseen paraphrases	v2 89.5% vs v1 71.0% over 38 held-out paraphrases (+18.4 pts); change-qa routing 0.0% to 100.0%	**supported**
4. Mask-derived change-qa vs a learned/generative head on CDVQA	on RGB change_or_not (n=3493): mask-derived 53.7% vs generative 59.2% (floor 54.2%) -- the mask method needs the spectral bands it has on the system's own tiles, absent from RGB CDVQA	**bounded**

Table 4 — Ablation outcomes. Claims 1-3 are supported on measured data; claim 4 is honestly bounded (the mask method's edge is in the multispectral setting, not on RGB-only CDVQA).

Ablation 1 detail — water recall vs cloud cover

Cloud cover	Optical-only IoU	Fusion IoU	Optical recall	Fusion recall
0%	1.000	**1.000**	1.000	**1.000**
20%	0.629	**0.836**	0.629	**0.889**
40%	0.444	**0.701**	0.444	**0.777**
60%	0.363	**0.667**	0.363	**0.754**
80%	0.166	**0.538**	0.166	**0.639**

Table 5 — As synthetic cloud occludes the optical tile, optical-only water detection collapses while SAR-backed fusion (radar fills the clouded pixels) holds. Ground truth is clean-optical NDWI water, so both read 1.000 at 0% cloud by construction.

These four ablations were run on 5400 EuroSAT test tiles, five SAR sites, 3,493 CDVQA change questions and 38 held-out router paraphrases; the harness and raw JSON ship in the repo (scripts/run_ablations.py, data/ablation_metrics.json) so every number reproduces.

CONFIDENCE CALIBRATION (DESIGN SECTION 6.1)

Stage 3 temperature-scales each answer's confidence so that a reported 0.8 means 0.8. scripts/fit_calibration.py fits the temperature T on a held-out half of each benchmark (p_cal = sigmoid(logit(p)/T)) and measures Expected Calibration Error (ECE) on the disjoint half; results are read from data/calibration_metrics.json. The AUROC column asks a prior question: does a higher confidence actually predict a correct answer? (0.5 = the confidence carries no such information.)

Scorer / benchmark	Conf. AUROC	ECE raw	ECE calibrated	T
Zero-shot — RSVQA-LR	0.591	0.387	**0.118**	≥20 (pinned)
Zero-shot — VRSBench	0.527	0.450	**0.232**	≥20 (pinned)
Zero-shot — CDVQA	0.482	0.379	**0.032**	≥20 (pinned)
Adapted — RSVQA-LR	0.902	0.044	**0.021**	1.46
Adapted — VRSBench	0.814	0.066	**0.021**	2.33
Adapted — CDVQA	0.774	0.051	**0.029**	1.18

Table 6 — Calibration of the answer scorer's confidence. The deployed zero-shot scorer's confidence is near-uninformative on foreign benchmarks (AUROC ≈ 0.5), so a single temperature can only lower ECE by shrinking every confidence toward the base rate and T pins at the search ceiling. The task-adapted scorer (the mission-corpus deployment setting) has informative confidence (AUROC 0.77–0.90) and temperature scaling roughly halves its ECE at a sensible T.

This is why the pipeline does not surface raw softmax as its answer confidence: a zero-shot scorer that is 98% confident is right only ~47% of the time. Stage 2 instead grounds confidence in auditable spectral fractions and cross-model reconciliation, and Stage 3 applies the fitted temperatures for the tasks the adapted scorer drives (vqa T=2.11, change-qa T=1.18) with conservative priors elsewhere. Every number reproduces from scripts/fit_calibration.py.

CAPTION & GROUNDING COVERAGE (VRSBENCH)

The VQA numbers above are one of VRSBench's three tasks. The other two — free-form captioning and referring-object grounding — probe capabilities the deployed heads were not built for, and are reported here honestly rather than omitted. Metrics are computed with a dependency-free implementation (scripts/vlm_metrics.py); the system caption is produced by the deployed CaptionHead on each image.

Caption metric	BLEU-1	BLEU-4	ROUGE-L	CIDEr-D	n
SatQuery CaptionHead vs GT	0.043	0.000	0.059	0.002	9,350

Table 7 — VRSBench captioning. The CaptionHead is a BigEarthNet-19 land-cover describer (~27 words); VRSBench captions are free-form scene prose (~50 words). The low n-gram overlap is a genre/task mismatch, a lower bound on a caption capability never trained for — the same candid story as the zero-shot VQA column.

Grounding is a different task again: VRSBench grounds 26 discrete man-made object classes (vehicle, ship, airplane, tennis-court, …) while the system grounds areal spectral land-cover classes (water, vegetation, built-up, …). Vocabulary overlap is 0 of 16159 objects, so referring-object IoU is not applicable by construction — the honest statement, not a forced mapping. The IoU implementation itself is validated (self-check 0.1429) and is exercised on real masks by the fusion and change ablations above.

OPTICAL + SAR FUSION — PER-CLASS IOU (BIGEARTHNET)

The fuse intent (design section 5.3) claims that combining the optical and SAR land-cover heads reads a scene better than either sensor alone. This measures it directly on the held-out, manually verified BigEarthNet.txt bench split — 1,020 patches that carry BOTH an S2 optical and an S1 (VV/VH) SAR patch and were never used to fit the encoder or heads. Prediction is multi-label over the 19 BigEarthNet-19 classes, reproducing the deployed heads exactly (satquery.stage2.neural.lulc / .sar_lulc); fusion is late fusion of the two probability vectors. The metric is per-class IoU (Jaccard = TP / (TP+FP+FN) across the held-out patches), at the deployed 0.5 threshold. Numbers are read from data/ben_fuse_iou_metrics.json.

Setting	macro-mIoU	micro-IoU	macro-F1	example-IoU
Optical-only	0.295	0.396	0.418	**0.531**
SAR-only	0.257	0.347	0.376	0.439
Fusion (mean)	0.282	0.398	0.400	0.509
Fusion (max)	**0.322**	**0.411**	**0.454**	0.507

Table 8 — Per-class IoU on the held-out BigEarthNet bench split. Max-fusion of the two trained heads gives the best per-class IoU at the deployed threshold (macro-mIoU 0.322 vs 0.295 best single sensor, +0.027), because the SAR head rescues classes where the optical head sits just under 0.5.

The gain is exactly where radar physics predicts: fusion helps arable land (0.40→0.61), industrial or commercial units (0.03→0.16), land principally occupied by agriculture (0.10→0.22), urban fabric (0.33→0.44) — water (specular dark returns), built-up and industrial (strong structural backscatter) and arable land (surface roughness and moisture) — while it slightly hurts the spectral forest classes, where optical multispectral is the right sensor and radar adds noise. This mirrors ablation 1: SAR's decisive value is as a cloud-robust fallback, not an additive signal on clean scenes. Once per-class thresholds are tuned on the train split, a well-operated optical-only head

recovers most of that (macro-mIoU 0.351) and edges past fusion — SAR's land-cover signal is largely redundant with optical when optical is clean and properly thresholded.

CHANGE-MASK LOCALISATION (LEVIR-CD)

The CDVQA results score change *questions*; this scores the change *mask* the deployed ChangeHead actually draws, against pixel ground truth. LEVIR-CD test (2,048 co-registered 256×256 bi-temporal tile pairs, 5.1% change pixels) carries binary **building**-change masks. The ChangeHead is unsupervised change-vector analysis (per-pixel spectral L2 difference above a fraction of the scene 90th percentile); metrics are for the change class over all pixels, read from data/levir_cd_metrics.json.

Operating point	Precision	Recall	F1	IoU
Deployed (cva_thresh 1.5)	0.047	0.441	0.085	0.044
Best-F1 sweep (cva_thresh 0.30)	0.050	0.927	**0.096**	**0.050**

Table 9 — LEVIR-CD change-mask scores. Recall is high (the change regions do fire) but precision sits near the 5% change-pixel base rate: unsupervised CVA flags all spectral change — seasonal, vegetation, illumination — and on RGB it cannot separate those from building change. This is a lower bound on unsupervised RGB change localisation, not a building-change detector.

The reading is the same as ablation 4: the change pipeline's edge is in the multispectral setting. On the system's own Sentinel-2 / Sentinel-1 tiles the ChangeHead works from per-class spectral deltas (NDWI / NDVI / NDBI) and the SAR-backed fusion path — where ablation 1 measures it holding 64% water recall under 80% cloud — not from raw three-band CVA against a building-only mask. LEVIR-CD is RGB with no such bands, so it bounds the degraded case honestly.

RSVQA-HR — HIGH-RESOLUTION AERIAL VQA

RSVQA-HR (Zenodo 6344367; 15 cm USGS aerial, 512×512 tiles) is the high-resolution companion to RSVQA-LR above. It is evaluated with the same scorer and protocol — zero-shot (the BigEarthNet.txt joint scorer applied directly) and, when shown, supervised (adapted on the HR train split). Count and area answers are binned into the paper's five ranges before scoring; presence and comparison are yes/no. Numbers are read from data/rsvqa_hr_metrics.json.

Question type	Zero-shot	Supervised
Presence	59.50%	**90.38%**
Comparison	49.54%	**81.68%**
Counting (binned)	32.82%	**86.45%**
Area (binned)	3.95%	**84.28%**
Overall	**41.02%**	**85.60%**

Table 10 — RSVQA-HR accuracy by question type (test set 1, 222,684 scored questions). Supervised adaptation lifts overall accuracy from 41.0% to 85.6%.